

Analyzing Voter Trends During the 2020 Election in San Francisco Using De-Randomized Ballot Data

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1. Introduction

Following the 2020 election, there was extensive security research done to ensure the security of elections. Through this security research, vulnerabilities that betrayed voter privacy were discovered, including the DVSorder vulnerability. This vulnerability stems from scanners made by Dominion Voting Systems, which anonymize ballot level data using an insecure technique. Dominion machines store a digital representation of each ballot in a file called a cast vote record (CVR). To anonymize the data, they use a pseudo-random number generator and various obfuscations to scramble up the ballots in each CVR file, preventing people from deducing what order the ballots were cast in. However, the DVSorder vulnerability showed that it is possible to unscramble ballots (using just CVRs). As a result, votes could be sorted in the order they were cast. Since many jurisdictions were under the impression that the data was anonymized, some CVR data has been published online for the public to access. One such jurisdiction, San Francisco, also releases scanner logs from their Dominion Systems machines.

Using DVSorder to unscramble the ballots, and timestamped scanner log files, we reconstructed a timeline of the 2020 election in San Francisco to analyze voter trends. We studied different races in the election, such as the Presidential, US House, California Senate, and local contests. Along with the individual contests, we looked at overvotes, undervotes, and voting across precincts, searching for trends throughout the day. By constructing a timeline of the votes, we are able to identify patterns in the voting data and reveal how voter privacy is compromised.

2. Data and Methodology

In order to create a timeline of the votes cast, we had to find published CVR data along with log files, specifically from vulnerable Dominion systems. This requirement made many sources of data void because they did not fall into both categories.

2.1 - Discovery

To study voter behavior, we first had to find voter data. We had two requirements for data. First, it had to come from a vulnerable Dominion machine: the ImageCast Evolution (ICE) or the ImageCast Precinct (ICP). Second, it had to be full CVR data. This was difficult because while these are common tabulator models, few jurisdictions release CVR data. We found many jurisdictions that release CVR data that don't use an ICE or ICP, and many jurisdictions that use an ICE or ICP but don't release CVR data. We found data from San Francisco in 2020, 2022, and 2024, and the state of Alaska in 2022 and 2024.

2.2 - Exploration

Next, we explored the CVR data and log files. We found that each San Francisco log files' title was associated with a unique tabulator ID, allowing us to associate logs with CVRs. We decided to focus mostly on San Francisco instead of Alaska, as the logs released by SF gave us the unique opportunity to create a specific timeline. We also reached out to Walter Mebane, professor of political science and statistics at the University of Michigan. Professor Mebane provided us with relevant social science papers whose findings we could attempt to reproduce with our data.

2.3 - Manipulation

We used a tool created by J. Alex Halderman to unscramble the ballots, giving us a list of ordered ballots (each ballot has a unique ID number) for each tabulator (each tabulator also has a unique ID number).

2.4 - Processing

Once we had the ballots in sorted order, we added timestamps to them. To do so, we parsed each log file. Logs contain detailed records of every action the tabulator takes, with timestamps. We noted a line in the logs that read “[ScannerThread] INFO : [Scanner] Ballot cast”. We stored instances of this line from the bottom up (going top-down presented issues such as test ballots being cast). We also took note of a line near the bottom that read “INFO : [Counters] Number of ballots cast:”. We then took the timestamp from each line and associated with the relevant ballot from the unscrambled ballots (for example, the earliest timestamp was associated with the first ballot cast, next timestamp with the second ballot cast, and so on). We then had a list of sorted ballots with timestamps partitioned by their tabulator ID. After this, we were able to add the timestamps to the relevant CVRs, easily indexing by tabulator and ballot ID numbers. Along the way, we made sure to check for errors, including comparing the number of ballots cast reported by the log file with the number reported by the DVSorder tool. We ended up discarding 34/587 CVRs due to errors, giving us a dataset of 553 CVRs.

2.5 - Analysis

Once we had sorted CVRs with timestamps, we graphed the data to visualize any trends. We also looked into revealing any specific voter privacy violations.

2.6 - Interpretation

After graphing, we examined the data to make key takeaways and compare with social science findings.

3. Results and Findings

3.1 - Presidential Contest San Francisco 2020

The Presidential Election in San Francisco was hardly close in the 2020 election. However, this is a trend that many have seen in election results over the past many years. California in general, and San Francisco, vote overwhelmingly Democratic. One lesser known trend is that so many more voters come in the morning hours, and the number continues to decrease as the day goes on. Even at the end of the work day there is no significant increase in votes, which would suggest people come following work on Election Day. Instead, there is actually a decrease in voter turnout after 5pm. This trend of early voting on Election Day is one with evidence from multiple contests in San Francisco.

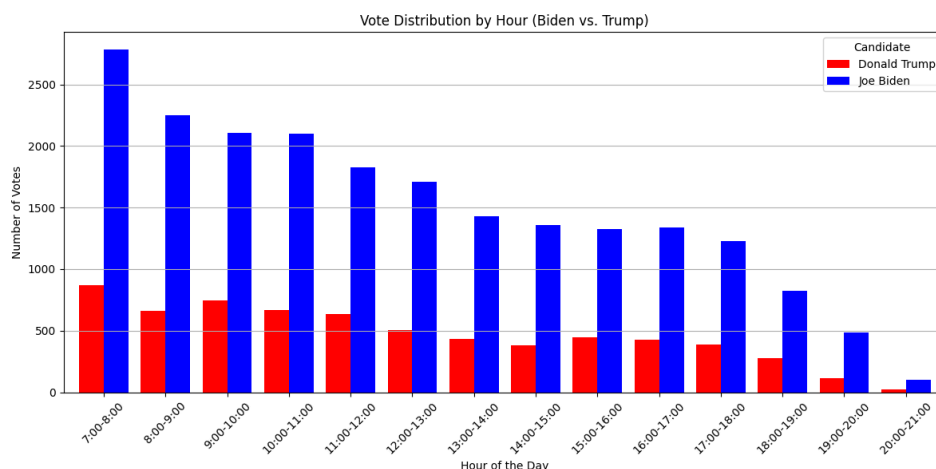


Figure 1: Vote distribution by hour for the 2020 Presidential Election in San Francisco

3.2 - US House Contest San Francisco 2020

For the US House elections we focused on two key districts within San Francisco, Districts 12 and 14. District 12 includes the Oakland metropolitan area and District 14 includes the suburbs of Alameda.

The same trends that were noticed in the Presidential Election held once again, with a large majority of voters from either party voting earlier in the morning. One fairly interesting observation is the number of votes per hour between District 12 and District 14 in the US House Election. In District 12, the total number of votes per hour varied from 1,000 to 2,000 votes, while District 14 ranged from 50-200. One reason for this difference is the makeup of these two districts, with District 12 being more of an urban center versus 14 being more suburban. This shows the population difference between the two districts, but also calls for greater analysis on the turnout of suburban voters in the US House Election to see if it varies from urban centers.

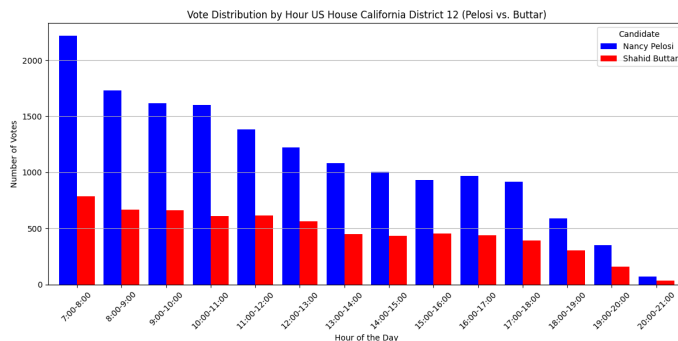


Figure 2: Vote distribution by hour for the 2020 US House Election District 12 in San Francisco

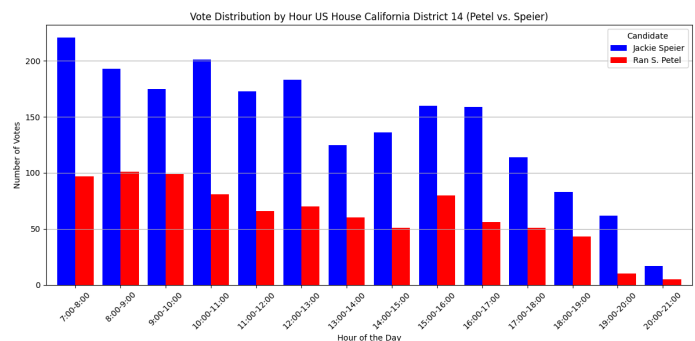


Figure 3: Vote distribution by hour for the 2020 US House Election District 14 in San Francisco

3.3 - Overvotes

According to the U.S. Election Assistance Commission (EAC), overvotes occur “when a voter makes more than the permitted number of selections in a single race/contest or when a voter makes a selection in a race/contest on which he/she was not eligible to vote”. By leveraging reconstructed ballot data and scanner logs, we examined overvote patterns in the San Francisco 2020 election across different races and time periods, focusing on identifying patterns and potential causes.

Overvote Trends by Hour

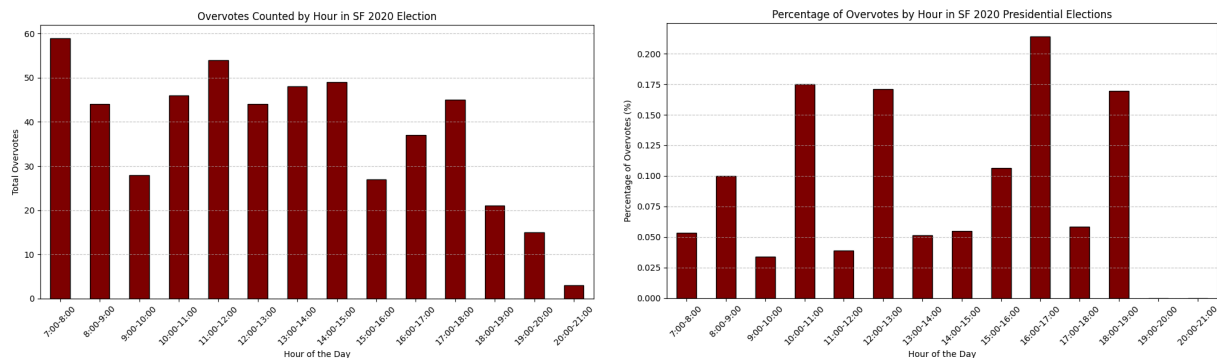


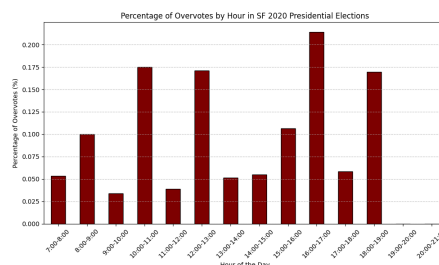
Figure 4: Overvotes trends by hour in San Francisco 2020 election

We were able to reconstruct a timeline of overvote trends throughout election day using the timestamped CVR data. However, based on our observations, overvotes appear to occur at random as we saw no clear pattern when analyzing the percentage of voted contests that contain overvotes. For example, peaks in overvote percentages are observed during mid-morning (10:00–11:00 AM), early afternoon (2:00–3:00 PM), and evening (7:00–8:00 PM), with no consistent correlation to voter turnout trends. In contrast, when looking at the total number of overvotes, trends align more predictably with voter turnout. Total overvotes are highest in the early morning hours (7:00–8:00 AM) and gradually decline as the day progresses, reflecting the overall decline in voter turnout. This disconnect between the total number of overvotes and the percentage of contests with overvotes underscores the randomness in overvote occurrences across different contests and times of day. These observations suggest that overvotes are not driven by consistent factors tied to specific voting periods but may instead reflect individual voter behavior and ballot design issues across the contests.

Overvote Trends Across Races

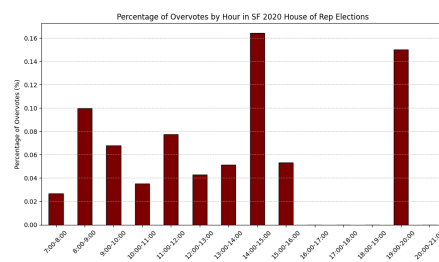
Presidential Contest:

The Presidential election, despite being a high-profile race, showed scattered overvote percentages. Overvotes peaked in the late morning (10:00–11:00 AM), mid-afternoon (12:00–2:00 PM and 4:00–5:00 PM), and evening (6:00–7:00 PM), reaching a maximum of approximately 0.2% during these times. The sporadic nature of these peaks, unaligned with voter turnout trends, underscores the random nature of overvote occurrences in this contest.



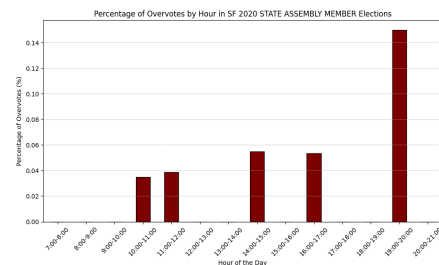
House of Representatives Contest:

In the House of Representatives election, overvote percentages also demonstrated an erratic distribution. Peaks occurred at 8:00–9:00 AM, 2:00–3:00 PM, and 7:00–8:00 PM, with the highest percentage reaching nearly 0.16% in the evening. Despite the overall low percentages of overvotes in this contest, the uneven spikes throughout the day point to a lack of consistency, further suggesting randomness in their distribution.



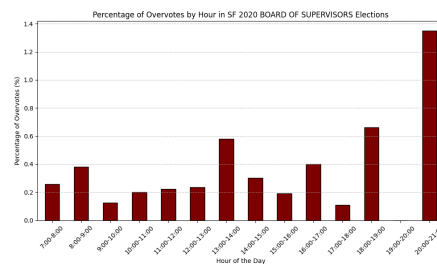
State Assembly Member Contest:

In the State Assembly Member election, overvotes were sparse throughout most of the day, with percentages remaining near zero during the majority of voting hours. However, there was a sharp and isolated spike at 7:00–8:00 PM, where overvote percentages exceeded 0.14%. This dramatic increase, occurring solely during the final voting hour in which we observed low voter turnout, highlights the irregularity of overvote distribution, reinforcing the randomness of their occurrence across the timeline.



Board of Supervisors Contest:

Overvotes in the Board of Supervisors election exhibited slightly more fluctuation throughout the day. Percentages remained relatively low during the early morning and midday hours but saw notable increases at 1:00–2:00 PM, 6:00–7:00 PM, and 8:00–9:00 PM, peaking at over 1.2% during the final voting hour. These varying peaks, uncorrelated with overall voting trends, suggest that overvotes for this contest do not follow predictable patterns but instead appear sporadically.



3.5 - Undervotes

According to the U.S. Election Assistance Commission (EAC), undervotes occur “when a voter makes less than that allowed number of selections in a single race/contest or when a voter votes on less than all of the races/contests for which he/she is eligible to vote.” By examining reconstructed ballot data and scanner logs from the San Francisco 2020 election, we identified patterns in undervote behavior across various races and time periods to better understand their causes and implications.

Undervote Trends by Hour

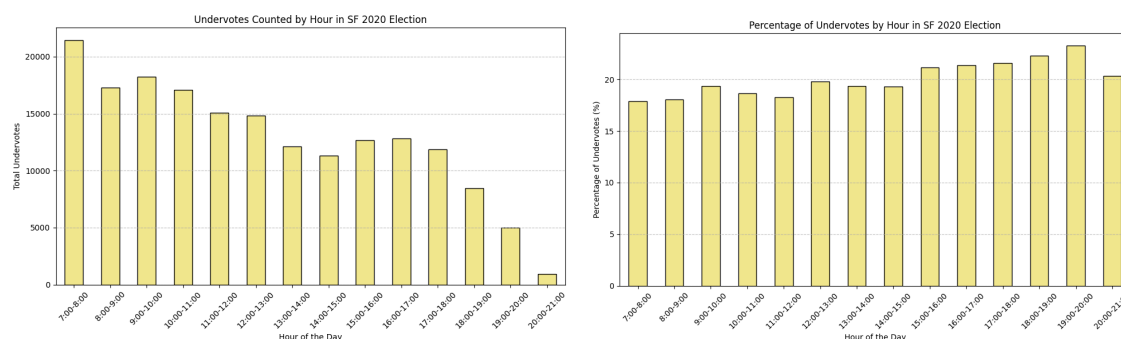


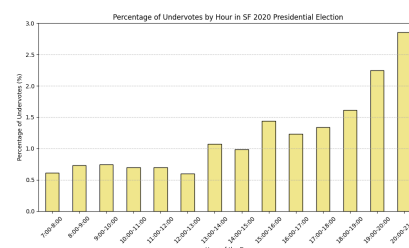
Figure 5: Undervote trends by hour in San Francisco 2020 election

Using the cast ballot data in the CVRs, we were able to analyze trends in undervotes throughout election day. The highest voter turnout occurred during the early morning hours (7:00-10:00 a.m.), resulting in the highest total number of undervotes cast during this time period. However, the percentage of undervotes in early morning contests was lower than at other times of the day. Midday voting (12:00-2:00 p.m.) had similar undervote percentages to the morning, but saw fewer total undervotes due to lower voter turnout. In the evening hours (7:00-9:00 PM), the percentage of undervotes in voted contests increased significantly, even though the total number of undervotes was lower, reflecting a higher proportion of skipped contests later in the day.

Undervote Trends Across Races

Presidential Contest:

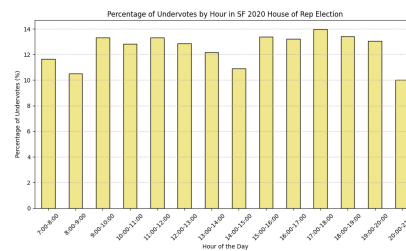
The presidential election had significantly lower undervote percentages than other contests, starting below 1% in the morning (7:00-8:00 AM) and remaining steady throughout most of the day. A gradual increase occurred in the late afternoon (4:00-5:00 p.m.), peaking at about 2.5% during the last hour of voting (8:00-9:00 p.m.). The low and steady



undervote percentages reflect high voter engagement and familiarity with the candidates in this high-profile, top-of-ballot race.

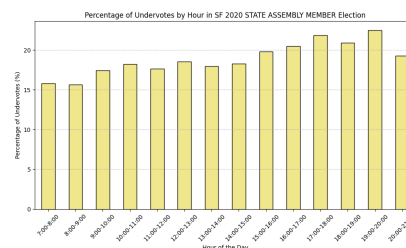
House of Representatives Contest:

Undervote percentages in the House of Representatives election were consistent throughout most of the day, starting at 10% in the morning (7:00-8:00 AM) and gradually increasing to 12-13% by midday. The percentages remained steady in the afternoon and evening, with a slight drop in the last hour of voting (8:00-9:00 PM). This contest had less volatility than others, with a relatively even distribution of undervotes throughout the day.



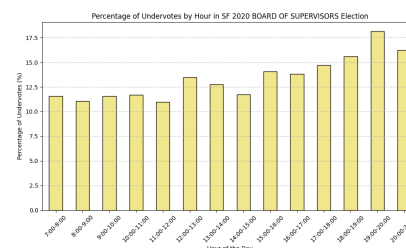
State Assembly Member Contest:

The undervote percentage in the State Assembly election started at about 15% in the early morning hours (7:00-8:00 AM) and remained stable throughout the day, fluctuating between 15% and 17%. A gradual increase was observed beginning in the late afternoon (3:00-4:00 p.m.), peaking at over 20% during the last hour of voting (8:00-9:00 p.m.). This trend highlights a steady increase in undervotes as the day progressed.



Board of Supervisors Contest:

The Board of Supervisors election had consistently higher undervote percentages compared to other contests, starting at around 12% in the morning (7:00-8:00 AM). The percentages remained steady throughout the noon hour, but increased in the late afternoon (3:00-4:00 p.m.) and peaked at over 17% during the final voting hour (8:00-9:00 p.m.). The upward trend throughout the day underscores the increased complexity of this contest, as it utilized ranked-choice voting.



3.6 - Voting Errors

As part of our research, we reached out to social science researchers to gather relevant studies on voter trends, aiming to compare our findings with existing work in the field. While we will discuss the challenges of this process in greater detail in the challenges section, we encountered difficulty comparing our data to most of the research we were directed toward because the datasets we analyzed lacked voter demographic information, whereas social science research predominantly focuses on demographic trends and their impact on voter behavior. However, Professor Walter Mebane from the University of Michigan provided a valuable resource that aligned with our study's focus. He pointed us to the Democratic National Committee's report, *Democracy at Risk: The 2004 Election in Ohio*, which indicated that 33% of voters in Ohio experienced errors before 8 a.m. and 21% of voters after 3 p.m. Inspired by this study, we decided to examine whether these trends held true 20 years later.

Using the combination of CVR and log data, we organized ballots from the 2020 San Francisco election into two groups: those cast before 8 AM and those cast after 3 PM. For each group, we counted the number of voters who experienced an error, and calculated the error percentage by dividing that number by the total number of valid ballots cast in each group. This analysis allowed us to determine the relative prevalence of errors during these time periods and to assess whether the trends observed in the *Democracy at Risk* report for the 2004 Ohio election held true in the context of the SF 2020 election.

Our results showed that, in contrast to the 2004 Ohio election, 38% of voters before 8 AM and 33% of voters after 3 PM experienced voting errors in the 2020 San Francisco election. Consistent with the DNC report, we found that early voters experienced significantly more errors than afternoon voters, although our results weren't as pronounced as those in the 2004 Ohio election.

3.7 - Voting Trends by Time

Voter Density by Time:

Using our timeline, we were able to analyze the density of votes by time. We see that the trend of more morning votes is again visible here, with most voters casting their vote between 7:10 and 7:30 AM. We also see it drops off steadily throughout the day, with some times after 8 PM having no votes cast at all.

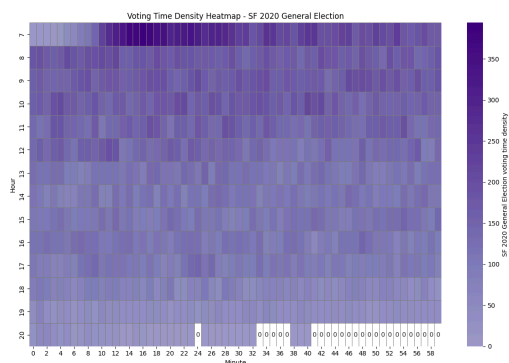


Figure 8: San Francisco voter density by time heatmap for 2020 election

Party-affiliated Votes by Time:

By computing a bias score of Democrat votes minus Republican votes at each timestamp, we are able to examine how party affiliation affects time of vote. However, since San Francisco votes overwhelmingly Democrat, this mostly just reinforces our finding that more voters vote in the morning. There is only one timestamp in the day where Republicans cast more votes than Democrats, which is 8:14 PM.

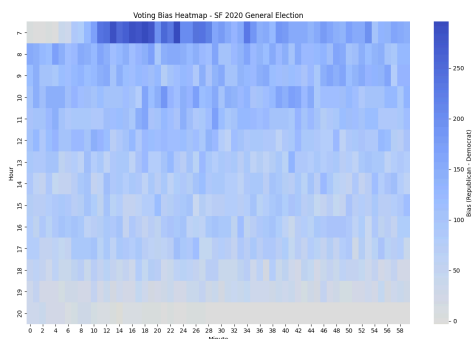


Figure 9: San Francisco voter bias by time heatmap for 2020 election

3.7 - Violating Specific Voter Privacy

The DVSorder vulnerability exposes a significant weakness in the way both the Dominion ImageCast Precinct and the Dominion ImageCast Evolution tabulators assign and obfuscate record IDs, undermining voter privacy. While Dominion assured its customers that their obfuscation methods made it safe to release CVR data and log files, the vulnerability allows for the reconstruction of voting order. Publicly available CVR data contains record IDs and vote choices, while log files provide detailed machine behavior information with timestamps. By combining these datasets, a timeline can be constructed that

assigns timestamps to ordered voter IDs. If a malicious actor had access to a camera at the polls or some other method of observing the order and time in which individuals voted, they could combine this information with the reconstructed timeline to determine how each observed voter cast their ballot. This poses serious risks to voter privacy, including the threat of coercion or vote buying, and fundamentally compromises the integrity of the election process.

Due to legal and ethical reasons, we did not follow this procedure to pair observed voters with their ballot choices. However, we were able to demonstrate the real-world impact of this vulnerability by analyzing a publicly shared social media post. By matching this publicly available information to the reconstructed timeline, we were able to show that voter privacy can indeed be compromised, underscoring the critical need to address this vulnerability to protect the integrity and anonymity of the voting process. We found a post on Twitter by a San Francisco Chronicle photographer that showed a voter at the voting booth. It also included a timestamp of when it was posted (8:09 AM PST), and the location of the voting site. Further digging revealed this photographer's photos in an article, which identified the voter by name. We were able to look tabulator was at that voting site, and from there simply had to look at the votes with a timestamp of 8:09 AM, which we could reasonably posit was the voter's.

Of course, there are potential problems with this methodology. It is possible that the Tweet was not posted immediately after the image was captured (the file had no metadata). It is also possible that we did not properly find the precinct ID—due to the Covid pandemic, some voting sites were in unusual places including this site, located in an LGBT leather bar called the SF Eagle. The idea still stands, though, as any social media posts like this could be utilized to violate voter privacy, especially if a malicious actor was monitoring social media on election day.

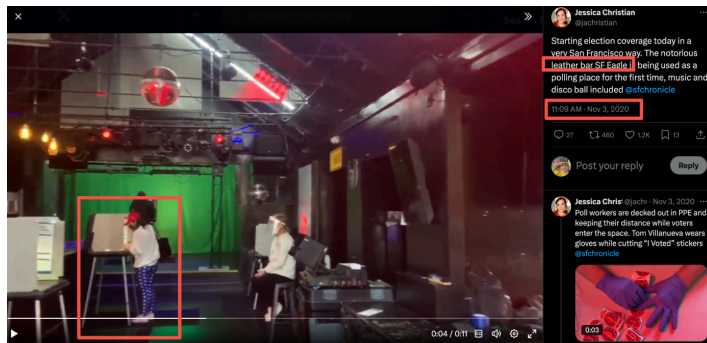


Figure 10: A 2020 Tweet showing a voter voting with a timestamp and location

4. Discussion

Our study of voter behavior based on reconstructed ballot data led to exploring social science research and discussing voter trends.

4.1 - Limitations

It is important to acknowledge that our dataset had some limitations. We had to omit 34 CVRs from the dataset due to errors. There were nine CVRs that had missing log files. There were 17 tabulators that could not be unscrambled with the DVSorder tool. There were seven CVRs whose log file vote counts did not match up with their DVSorder tool vote counts. There was one log file whose timestamps were irregular (votes starting in the evening and ending in the morning). We discarded any files with errors, but still had a robustly sized dataset of 553 CVRs and matching logs.

It is also important to note that the Covid-19 pandemic was still raging during the 2020 election. This may have impacted our data, especially because many voters may have chosen to mail in their votes rather than risking an in-person trip to the polls. Mail-in ballots are not vulnerable to the DVSSorder, so they are not represented in the data.

4.2 - Challenges

While the DVSSorder vulnerability is a significant issue that allows thorough investigation of voter behavior, gaining tangible insights from it requires access to both the vulnerable CVR data (from counties using Dominion ICP/ICE systems) and the corresponding log files. Unfortunately, the availability of these two critical components is extremely limited. San Francisco is the only county in the United States where we were able to obtain both vulnerable CVR data and log files, severely limiting the scope of our research. While we found vulnerable CVR data from the state of Alaska in 2022, the data was removed from their website after the 2024 election, and the 2024 data was not deemed vulnerable by the DVSSorder tool. This suggests that Alaska may have fixed the vulnerability. Without access to both required datasets from other jurisdictions (vulnerable CVRs + logs), the ability to fully investigate and analyze the impact of this vulnerability remains limited.

Additionally, our goal in analyzing voter trends was to connect our findings to existing social science research on voter behavior and trends across different states. However, a key difference lies in the type of data used. In our case, we had access to vote choice data but lacked demographic information, such as gender, ethnicity, or other identifiable traits. Conversely, social science researchers typically have access to demographic data but not individual vote choices. This fundamental difference made it challenging to directly compare or contrast our findings with existing social science research. Nevertheless, we believe that our observations could serve as a valuable resource for social science researchers in the future. If used ethically, integrating our findings with demographic data could help paint a more comprehensive and nuanced picture of voter behavior and decision-making processes.

This project presented the unique opportunity to study voter behavior in ways that have rarely, if ever, been possible before. However, because of this, we had to prioritize the data we deemed most important to study, which led to our focus on San Francisco county in the 2020 election. While this focus allowed us to produce meaningful insights, we acknowledge that prioritizing San Francisco, a highly liberal county, may introduce bias into the findings, as voter behavior and election trends in this area may not fully represent the diversity of political dynamics across the country. Despite this limitation, there is immense potential to build on this work by exploring voter behavior in other counties and other elections with vulnerable CVR data, even if log files are not currently available. This research represents just the beginning of what could be a much broader effort to better understand voter behavior and election integrity across the country. We are hopeful that this study will inspire others to continue investigating these important topics, leveraging the opportunities presented by this unique intersection of data and technology.

5. Conclusion

There were multiple overarching themes from the timelines generated from San Francisco's 2020 election data. In particular, voters tend to show up to the polls earlier in the day. This trend held for every contest in the 2020 election across party lines. This shows that voters prefer the morning to cast their ballot, which may stem from work or external obligations which prevent them from coming later in the day.

Furthermore, the trends in undervotes and overvotes reveal contrasting patterns that offer valuable insights into voter behavior and election dynamics. The percentages of undervotes demonstrate a consistent progression throughout the day, which suggests a shift in voter engagement or ballot completion as the day progresses. In contrast, overvote percentages displayed no discernible pattern, occurring seemingly at random throughout the day with peaks varying across contests. This lack of consistency in overvote trends demonstrates the complexity of identifying their root causes and highlights the importance of further research into ballot design, voting technology, and other potential factors that might contribute to these errors.

In addition, our voter error analysis revealed that early voters experienced higher error rates in both the 2004 Ohio election and the 2020 San Francisco election, suggesting a trend that warrants further investigation into the challenges faced during the early voting period. The higher overall error rates observed in San Francisco compared to Ohio may reflect differences in voting technology, ballot design, or election procedures, stressing the importance of more regular, localized studies to better understand and address these differences. These findings highlight the critical need to analyze voting trends over time so that researchers and policymakers can identify and mitigate persistent challenges in the election process, ultimately ensuring a more equitable and error-free voting experience for all.

Using our voting timeline, we could also examine voter density by time. We found the above trend of more morning voters to hold true, as vote density decreased throughout the day. We also looked at vote density by voter party. However, since San Francisco is so Democrat-leaning, we could not glean much useful information from this besides reinforcing the vote-in-the-morning finding. Future studies may find it interesting to normalize the voter bias to get more insight into voter trends by affiliation.

We also found that threats to individual voter privacy are very real. We were able to find a social media post with enough revealing information to possibly expose that voter's ballot. These social media attacks are possible, and may be even easier to execute if an attacker is actively monitoring social media on election day (as we believe many posts from the 2020 election have been lost to time by now). Additionally, an attacker with access to camera feeds (or one who could simply FOIA the surveillance tapes) could one-to-one match the votes cast with the voter who cast them.

Going forward, we believe that there is much more to be gleaned from data like this. We have not performed analysis on the 2022 or 2024 San Francisco data, nor the 2022 or 2024 Alaska data. If there were to be more jurisdictions to be found, they would provide interesting insight as well. This de-randomizable data, while a gem for researchers, is also a massive vulnerability for voters, and we call upon Dominion to rectify their obfuscation as soon as possible.

6. References

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